# Lecture 3.2

# Extended ER Modeling

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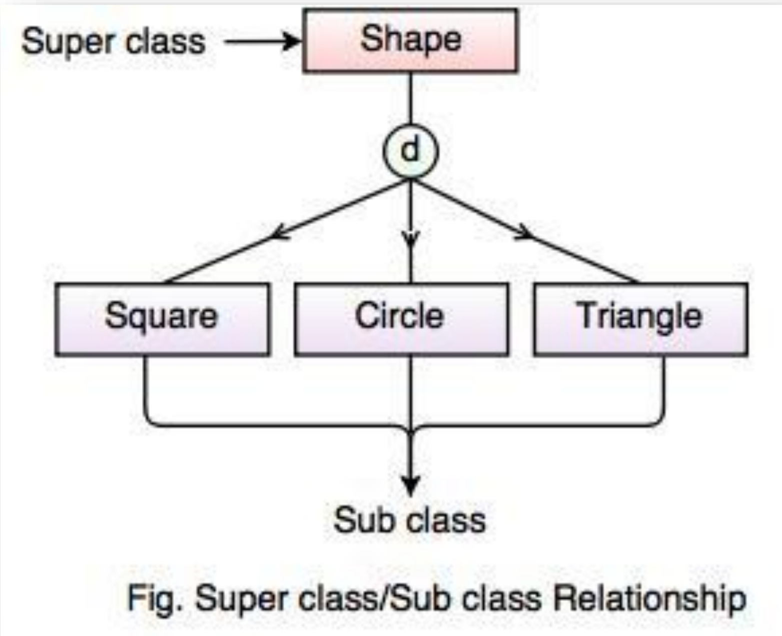
# Introduction

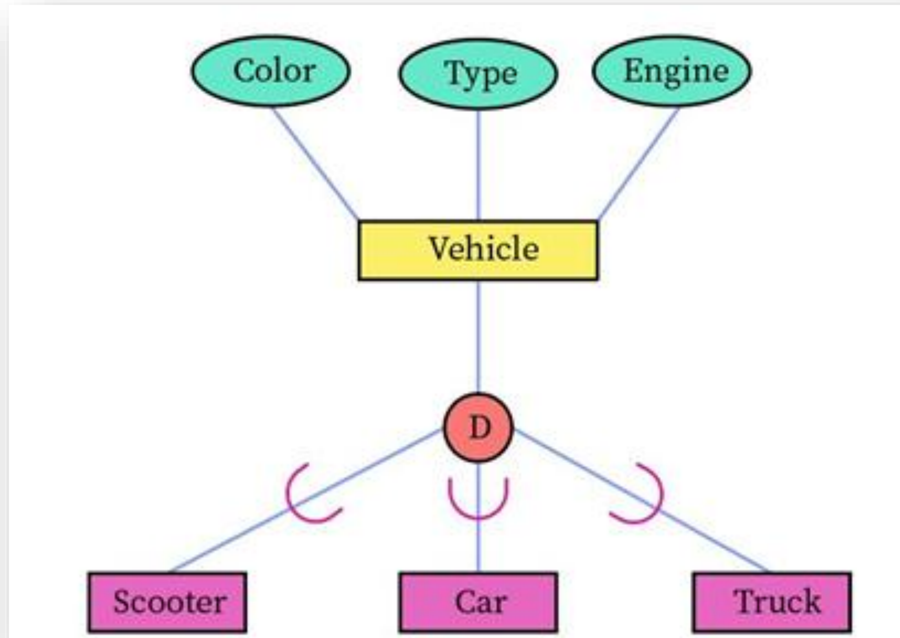
- The **original ER Model(1970)** was able to handle the **data modelling** of **the common business problems.**
- In **1980s,** there was a rapid increase in the development of many new database applications like **CAD(Computer Aided designing)**, **GIS(Geographical Information System)**, **Multimedia, Knowledge base for AI(Artificial Intelligence) database** and so on.
- The **basic ER-Modeling** concepts were **no longer sufficient** to represents the requirements of these **newer** and **complex applications.**
- The **ER-Model** that is supported with the **additional semantic concepts** is called **Extended ER Model(EER Model).**
- The **EER model** includes *all the modeling concepts of the **ER model***.
- In addition, it includes the concepts of **Subclass** and **Superclass** and the related concepts of **Specialization** and **Generalization.**

# Subclasses and Superclasses

- In some cases an **Entity type** has numerous additional **subgrouping** of its entities that are meaningful and need to be **represented explicitly** because of their significance to the database application.
- **Example**
- The entities that are members of the **EMPLOYEE entity type** may be distinguished further into *SECRETARY, ENGINEER, MANAGER, TECHNICIAN, SALARIED_EMPLOYEE, HOURLY_EMPLOYEE,* and so on.
- We call each of these subgroupings a **Subclass** or **Subtype** of the **EMPLOYEE entity** type, and the **EMPLOYEE entity** type is called the **Superclass** or **Supertype** for each of these **subclasses.**
- We call the **relationship** between a **Superclass** and any one of its **Subclasses** a **superclass/subclass** or **supertype/subtype** or simply **class/subclass relationship.**

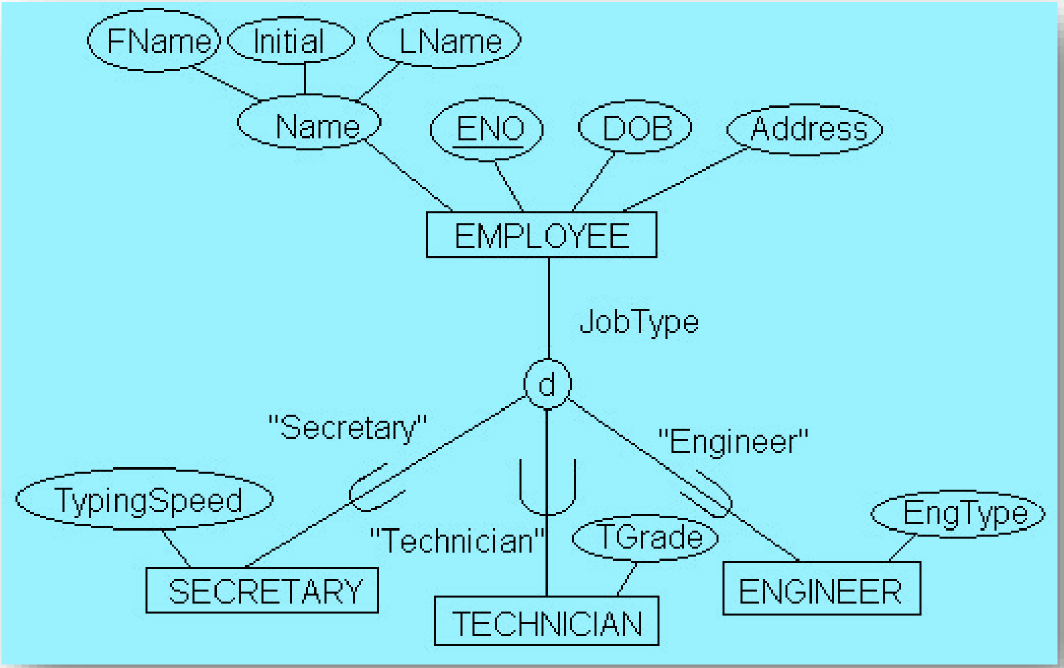
# Subclasses and Superclasses



# Attribute Inheritance

- **Attribute inheritance** is the property by which **subclass entities inherits all** the **attributes** of the **superclass entity.**
- This feature makes it **unnecessary to associate** the **superclass attributes** with the **subclass entities** thus **avoid redundancy.**



# Specialization

- **Specialization** is the **process** of defining a *set of **Subclasses*** of an **entity type**; this entity type is called the **Superclass** of the **Specialization.**
- **Specialization** is a **top-down process** of **maximizing the differences** between members of an entity by identifying their **distinguishing characteristics**.
- **Example,** the set of **subclasses** {*SECRETARY, ENGINEER, TECHNICIAN*} is a **specialization** of the **superclass EMPLOYEE** that distinguishes among employee entities based on the ***job type*** of each **employee entity.**
- We may have **several specializations** of the **same entity** type based on different **distinguishing characteristics.**
- **Example,** another **specialization** of the **EMPLOYEE entity type** may yield the set of subclasses {*SALARIED_EMPLOYEE, HOURLY_EMPLOYEE*}; this specialization distinguishes among employees based on the ***method of pay.***

# Representation of a Specialization diagrammatically in an EER

- The **Subclasses** that define a **Specialization** are **attached** by **lines** to a **circle** that represents the **specialization**, which is **connected i**n turn to the **Superclass.**
- The ***subset symbol*** on **each line** connecting a **subclass** to the **circle** indicates the **direction** of the **Superclass/Subclass** relationship.
- **Attributes** that apply only to entities of a particular **subclass**—such as *TypingSpeed of SECRETARY*—are attached to the rectangle representing that **subclass.**
- These are called **specific attributes** (or **local attributes**) of the **subclass.**

# Representation of a specialization diagrammatically in an EER

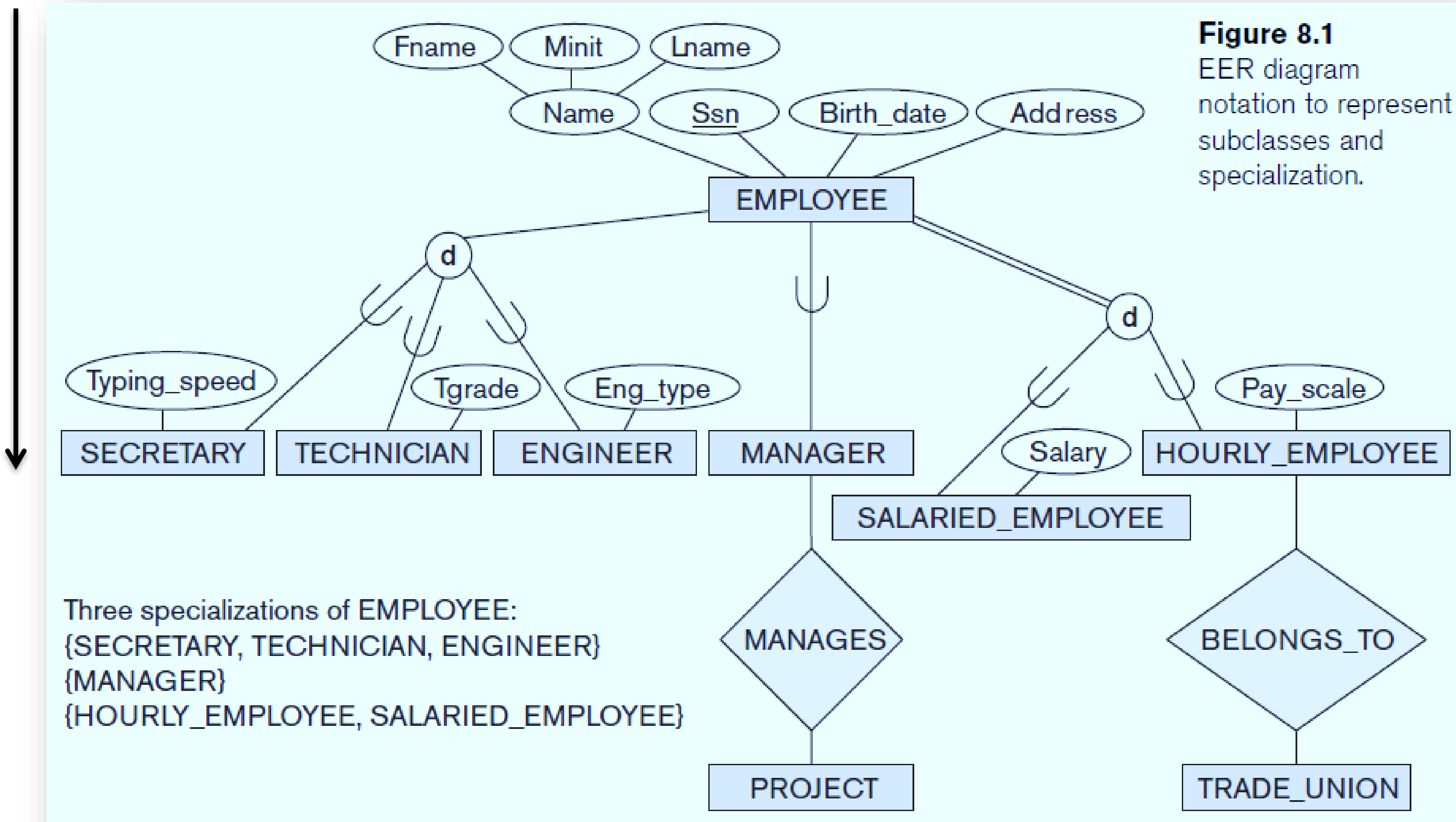
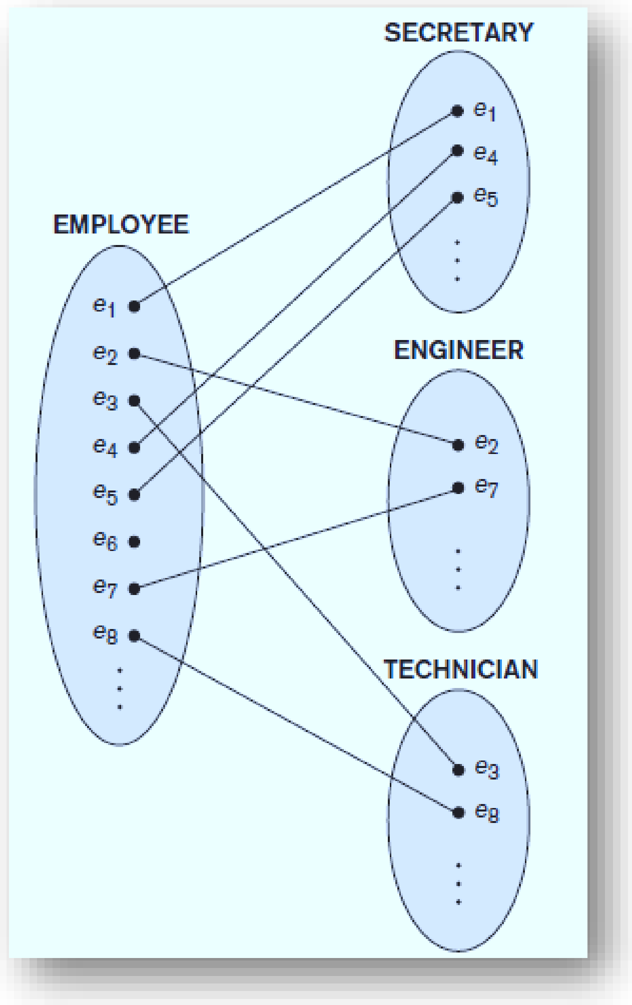



**Figure 8.1**
EER diagram notation to represent subclasses and specialization.

# Instances of a Specialization.

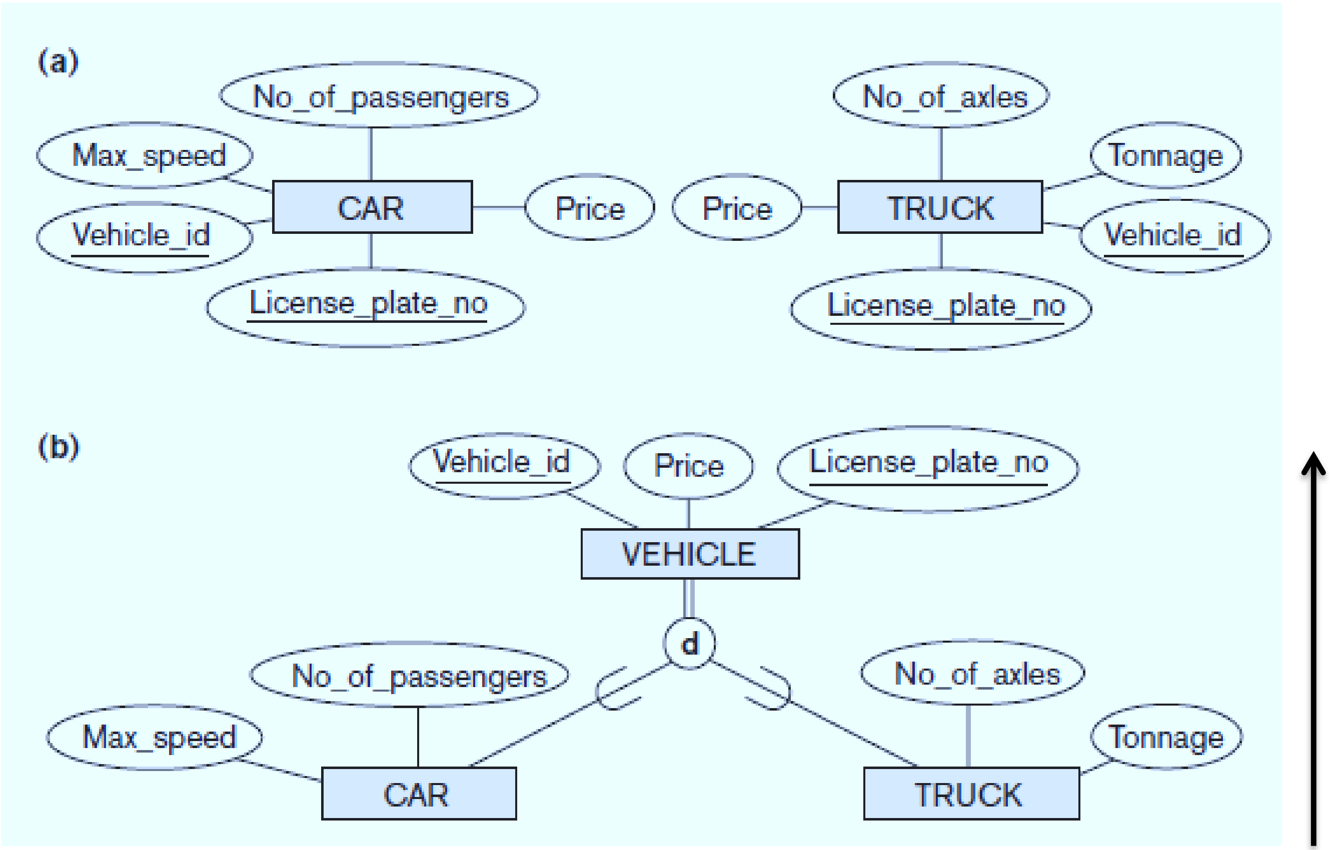


# Generalization

- **Generalization** refer to the process of **defining** a **generalized entity type** from the given entity types.
- **Generalization** is a **bottom-up process** of **minimizing the differences** between members of an entity by identifying their **common characteristics**.
- The **Generalization** process can be viewed as being **functionally the inverse** of the **Specialization** process.
- This approach results in the **identification** of a **generalized superclass** from the **original subclass.**
- **Example**, consider the **entity types CAR** and **TRUCK.**
- As they have **several common attributes**, they can be **generalized** into the **entity type VEHICLE.**
- Both **CAR** and **TRUCK** are now **subclasses** of the **generalized superclass** **VEHICLE.**

# Representation of a Generalization diagrammatically in an EER

- An **arrow** pointing to the **generalized superclass** represents a **generalization.**



# Constraints on Specialization and Generalization

- To model an enterprise more **accurately**, the database designer may choose to place certain **constraints** on a particular **Generalization/Specialization.**
   1. **Membership Constraints**
   2. **Disjoint Constraints**
   3. **Overlapping Constraints**
   4. **Completeness** (or **totalness**) **constraint**

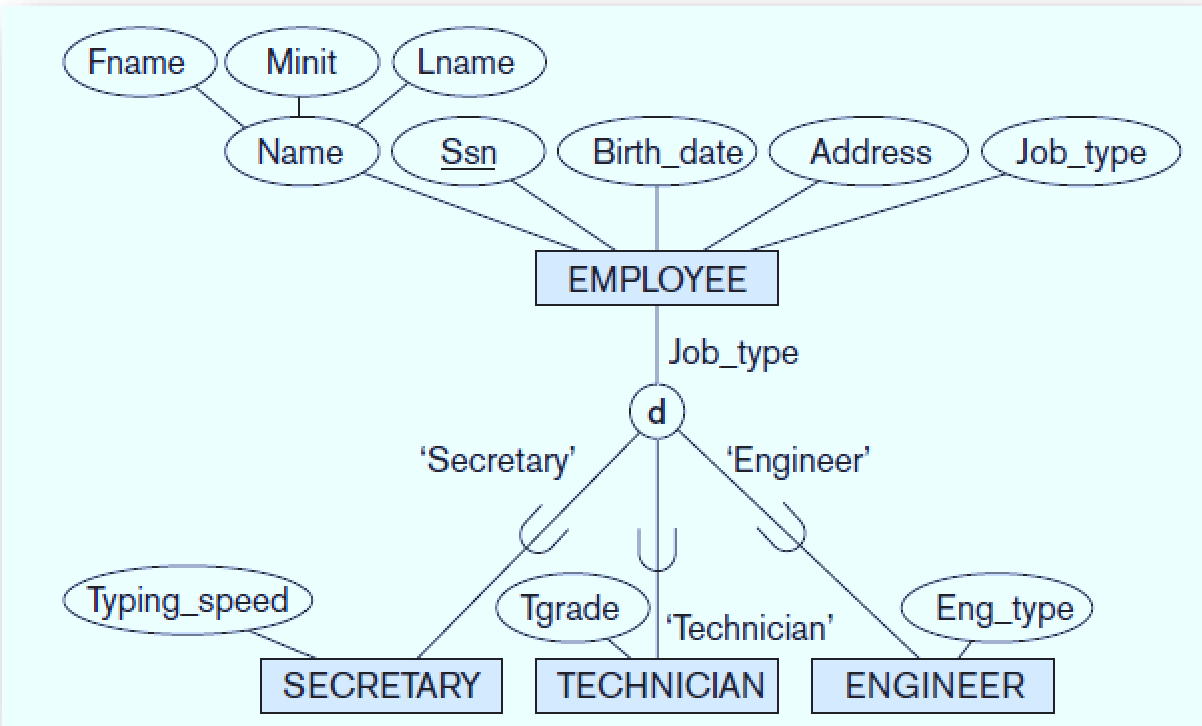
## 1. Membership Constraints

- This **constraint** is used for determining which **entities** can be **members** of a given **lower-level entity set.**
- In some **specializations** we can determine exactly the entities that will become members of each subclass by **placing a condition** on the **value** of some **attribute** of the **superclass.**

# Constraints on Specialization and Generalization

- Such **Subclasses** are called **predicate-defined** (or **condition-defined**) **subclasses**.
- **Example**, if the **EMPLOYEE** entity type has an **attribute** ***Job_type,*** we can specify the **condition** of **membership** in the **SECRETARY subclass** by the **condition** (**Job_type = 'Secretary**'), which we call the **defining predicate** of the **subclass.**
- This **condition** is a ***constraint*** specifying that exactly those entities of the **EMPLOYEE entity** type whose **attribute value** for **Job_type** is '**Secretary**' belong to the **subclass.**
- We display a **predicate-defined subclass** by writing the **predicate condition** next to the **line** that connects the **subclass** to the **specialization circle.**

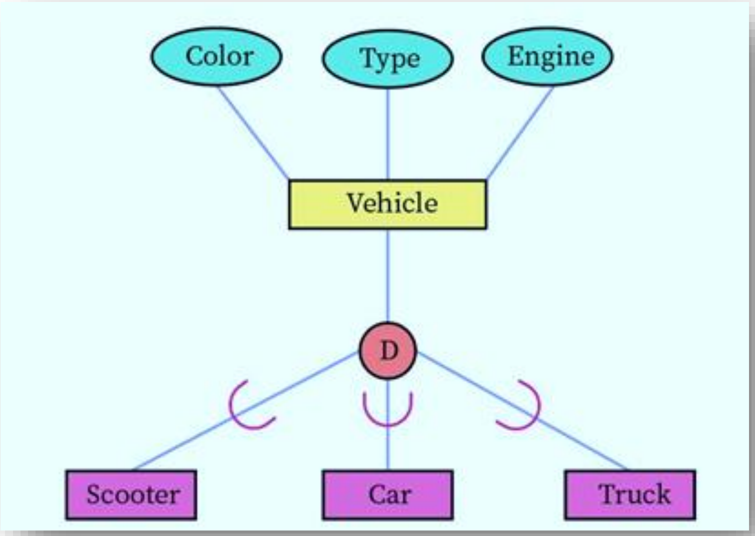
# Constraints on Specialization and Generalization



# Constraints on Specialization and Generalization

## 2. Disjoint constraint

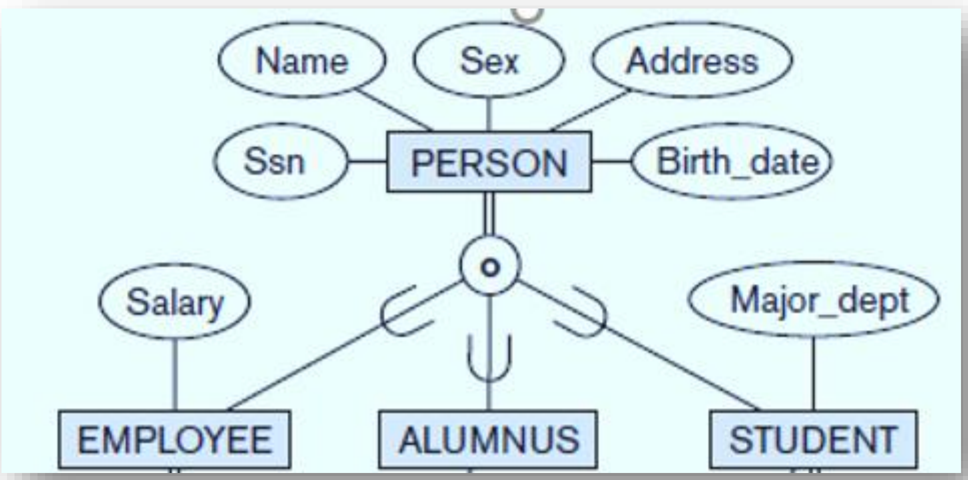
- This **constraint** specifies that the **subclasses** of the **specialization must be disjoint**.
- This means that **an entity** can be a member of **at most one** of the **subclasses** of the **specialization.**
- It is displayed by placing **d** in the **circle.**



# Constraints on Specialization and Generalization

## 3. Overlap constraint

- This **constraint** specifies that the **subclasses** of the **specialization** are **not constrained** to be **disjoint.**
- This means that an **entity** can be a **member of more then one subclasses** of the **specialization.**
- This case,is displayed by placing **O** in the **circle.**



# Constraints on Specialization and Generalization

**4. Completeness** (or **totalness**) **constraint:**

**(a). Total specialization**

**(b). Partial specialization**

**(a). Total specialization**

- **Total specialization** constraint specifies that ***every*** **entity** in the **superclass** must be a **member of at least one subclass** in the **specialization.**
- For **example,** if every EMPLOYEE must be either an HOURLY_EMPLOYEE or a SALARIED_EMPLOYEE, then the specialization {HOURLY_EMPLOYEE, SALARIED_EMPLOYEE} is a **total specialization** of EMPLOYEE.
- This is shown in **EER diagrams** by using a **double line** to connect the **superclass** to the **circle.**

# Constraints on Specialization and Generalization

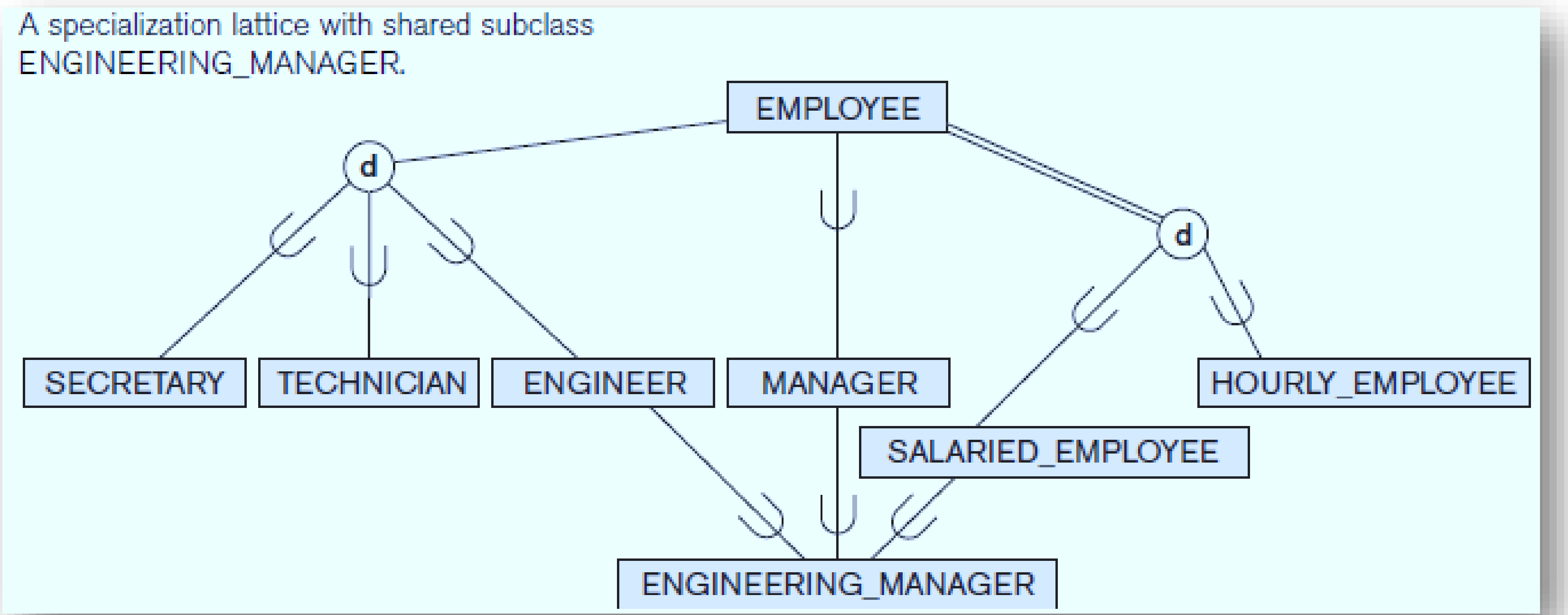
**(b). Partial specialization**

- **Partial specialization** constraint specifies that ***every*** **entity** in the **superclass** need not be a member of the subclass in the **specialization.**
- A **single line** is used to display a **partial specialization**, which allows an entity not to belong to any of the subclasses.
- For **example**, if some EMPLOYEE entities do not belong to any of the subclasses {SECRETARY, ENGINEER, TECHNICIAN}, then that specialization is **partial.**

# Specialization Hierarchies and Lattices

- A **subclass itself** may have **further subclasses** specified on it, forming a **hierarchy** or a **lattice of specializations**.
- For **example**, ENGINEER is a **subclass** of EMPLOYEE and is also a **superclass** of ENGINEERING_MANAGER.
- This represents the real-world constraint that every **engineering manager** is required to be an **engineer.**
- A **specialization hierarchy** has the **constraint** that each **subclass** has **only one parent**, which results in a **tree structure** or **strict hierarchy**.
- In contrast, for a **specialization lattice**, a **subclass** can have ***more than one*** **parent**.

# Specialization Lattice



# A specialization lattice with multiple inheritance

- In a **specialization lattice** or **hierarchy**, a **subclass inherits** the **attributes** not only of its **direct superclass**, but also of all its **predecessor superclasses** *all the way to the **root*** of the **hierarchy** or **lattice** if necessary.
- **Example**, an entity in **GRADUATE_STUDENT** inherits all the attributes of that entity as a **STUDENT** *and* as a **PERSON.**
- A **subclass** with ***more than one*** **superclass** is called a **shared subclass**, such as **ENGINEERING_MANAGER** and **STUDENT_ASSISTANT**.
- This leads to the concept known as **multiple inheritance**, where the **shared subclass** directly **inherits attributes** from **multiple classes**.

# A specialization lattice with multiple inheritance

- The **existence** of **at least one shared subclass** leads to a **lattice** (and hence to ***multiple inheritance*****).**
- If **no shared subclasses** existed, we would have a **hierarchy** rather than a **lattice** and only **single inheritance** would exist.

# A specialization lattice with multiple inheritance

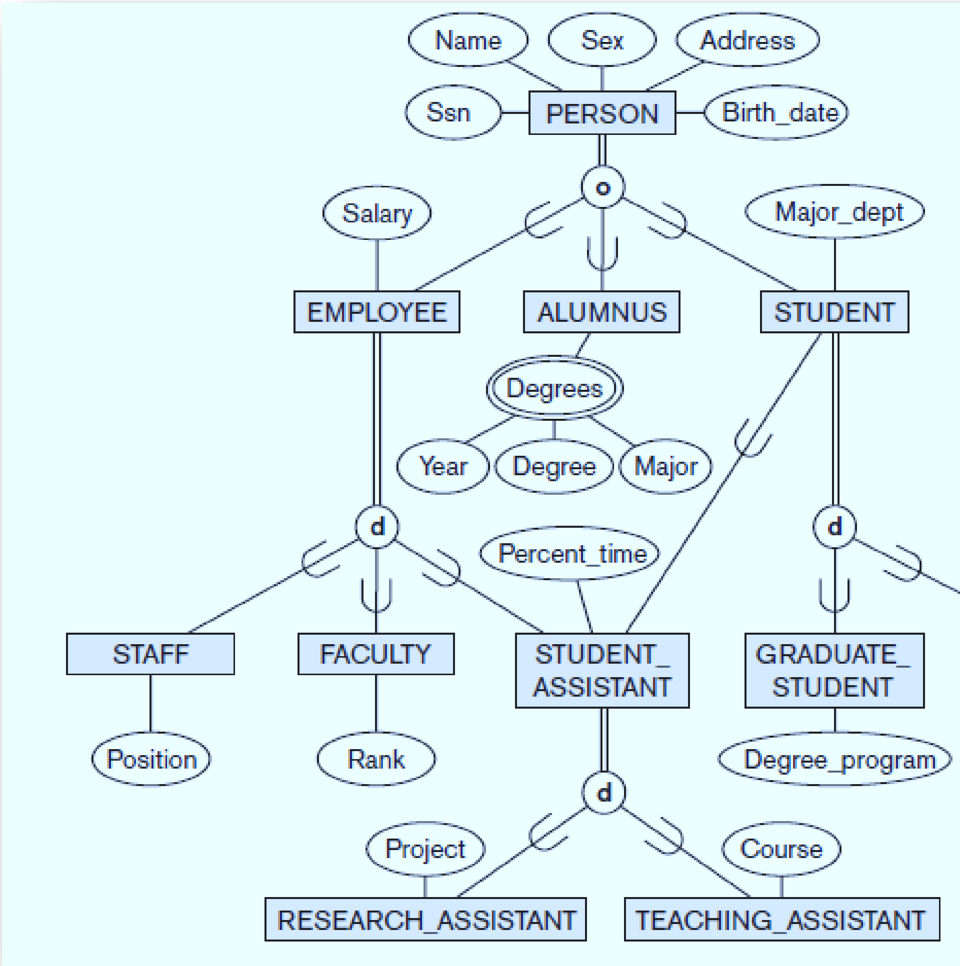
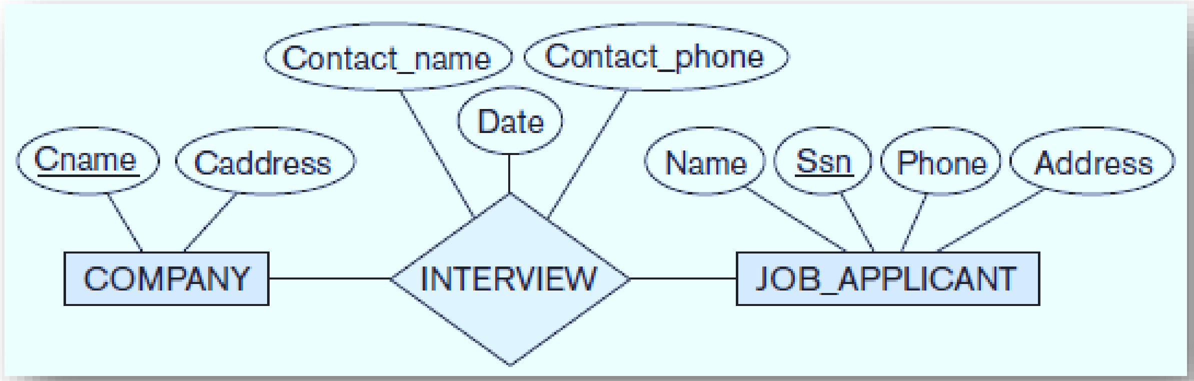



**Figure 8.7**
A specialization lattice with multiple inheritance for a UNIVERSITY database.

# Aggregation

- **Aggregation** is an **abstraction concept** for building **Composite objects(Entity set, Relationship Set)** from their **component objects.**
- Consider the **ER schema** shown in Figure, which stores information about **interviews** by **job applicants** to various companies.

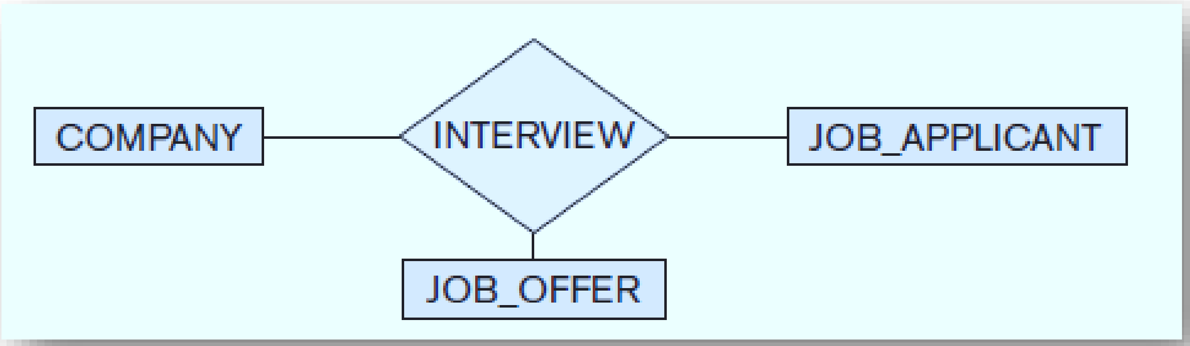



- The **relationship INTERVIEW attributes** *Contact_name* and *Contact_phone* represent the name and phone number of the **person** in the company who is responsible for the interview.

# Aggregation

- Suppose that **some interviews** result in **job offers**, whereas others do not.
- We would like to treat **INTERVIEW** as a relationship to associate it with **JOB_OFFER.**

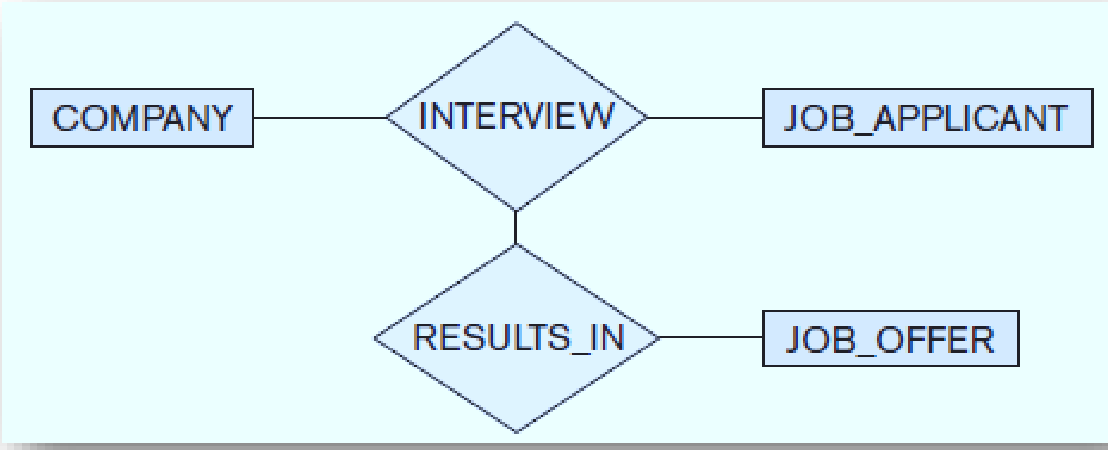
**Option 1**




- The **ER diagram** shown in Figure**: Option 1** is ***incorrect*** because it requires each interview relationship instance to have a job offer.

# Aggregation

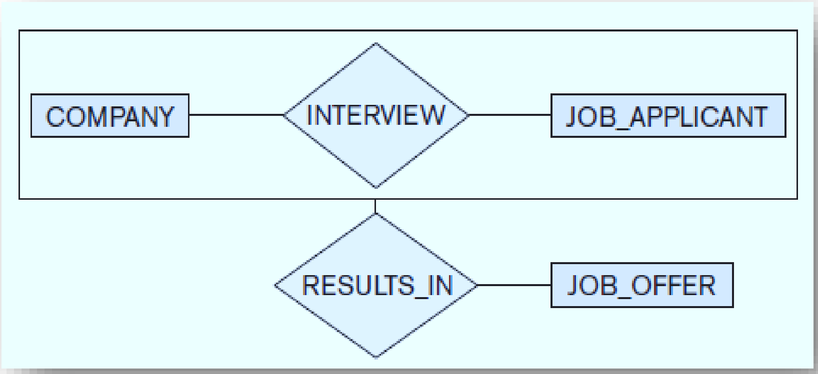
- **Option 2**




- The **ER diagram** shown in Figure(**Option2**) is ***not allowed*** because the **ER model** does **not allow** **relationships among relationships.**
- **Definition:** ***Aggregation*** *is an* ***abstraction*** *through which* ***relationships*** *are treated as* ***higher level entities.***

# Aggregation

- **Option 3**: Create a **higher-level Aggregate class** composed of COMPANY, JOB_APPLICANT, and INTERVIEW and to relate this class to JOB_OFFER.




- We regard the **relationship set** ***INTERVIEW*** (relating the entity sets *COMPANY and JOB_APPLICANT*) as a **higher-level Aggregated entity set.**
- Such an **entity set** is treated in the same manner as is any other entity set.
- We can then create a **binary relationship** ***RESULTS_IN*** between ***aggregated entity set*** and ***JOB_OFFER.***